

Cosets

Left Coset:

If $\{H, *\}$ is a subgroup of a group $\{G, *\}$, then the set aH , where $a \in G$, defined by

$$aH = \{a * h \mid h \in H\}$$

is called the left coset of H in G generated by the element $a \in G$. a is called the representative (element) of the left coset aH .

Right Coset:

Similarly the set H_a defined by

$$H_a = \{ h * a \mid h \in H \}$$

is called the right coset of H in G generated by $a \in G$. a is again called the representative (element) of H_a .

Example:

Let $(\mathbb{Z}_4, +_4)$ be the group w.r.t. Addition modulo 4.

where $Z_4 = \{[0], [1], [2], [3]\}$

& $H \subseteq Z_4$ where $H = \{[0], [2]\}$

be the subgroup of Z_4 w.r.t. $+_4$

for $0 \in Z_4$ $0 +_4 H = \{[0], [2]\}$ be
the left-coset

& $1 \in Z_4$ $H +_4 1 = \{[1], [3]\}$
be the right-coset

① Find the left cosets of $\{[0], [2]\}$ in the group $(\mathbb{Z}_4, +_4)$.

Sol:

Let $\mathbb{Z}_4 = \{[0], [1], [2], [3]\}$ be a group and $H = \{[0], [2]\}$ be a sub-group of $\mathbb{Z}_4$ under $+_4$ (addition mod 4).

$\therefore$ The left cosets of H are

$$[0] + H = \{[0], [2]\} = H;$$

$$[1] + H = \{[1], [3]\};$$

$$\begin{aligned} [2] + H &= \{[2], [4]\} = \{[2], [0]\} \\ &= \{[0], [2]\} \\ &= H \end{aligned}$$

$$\text{and } [3] + H = \{[3], [5]\} = \{[3], [1]\} \\ = \{[1], [3]\} \\ = [1] + H.$$

$$\therefore [0] + H = [2] + H = H \quad \text{and}$$

$[1] + H = [3] + H$ are the two distinct left cosets of H in Z_4 .

② Let $G = \{1, a, a^2, a^3\} (a^4 = 1)$ be a group and $H = \{1, a^2\}$ is a subgroup of G under multiplication. Find all the cosets of H .

Sol:

Let us find the right cosets of H in G .

Left Cosets

$$H1 = \{1, a^2\} = H$$

$$Ha = \{a, a^3\}$$

$$Ha^2 = \{a^2, a^4\} = \{a^2, 1\} = H$$

$$Ha^3 = \{a^3, a^5\} = \{a^3, a\} = Ha$$

$$\therefore H \cdot 1 = H = Ha^2 = \{1, a^2\} \text{ and}$$

$$Ha = Ha^3 = \{a, a^3\} \text{ are two}$$

distinct right cosets of H in G .

Similarly, we can find the left cosets of H in G .

Note:

- ① The element (a) is called the representative element of the left coset aH .

Ex:

$$\text{for } \mathbb{Z}_{12} / H = \{ [1], [3] \}$$

" $[1]$ " is the representative element.

- ② Let H be a subgroup of a group G . Let $a, b \in G$ if $b \in aH$ then $bH = aH$.

Ex:

for $Z_4 = \{0, 1, 2, 3\}$ & $H = \{0, 2\} \subseteq Z_4$
w.r.t. $'+_4'$.

Since $0 +_4 H = \{0, 2\}$ for $0 \in Z_4$

$1 +_4 H = \{1, 3\}$ for $1 \in Z_4$

Also $2 +_4 H = \{2, 0\} = 0 +_4 H$

Since $2 \in Z_4$ $2 \in 0 +_4 H$.

(3) Let H be a subgroup of G
group G then the set of all
left coset of H in G forms
a partition of G .

If any two distinct left cosets are disjoint
then the set of all (distinct) left cosets of H in G
forms a partition of G .

④ Let H be a subgroup of a group G then there is a bijection between any two left cosets of H in G . $\{ f : gH \rightarrow shH \}$
 bijective

⑤ The subgroup H itself is a left coset.

⑥ Every left coset of H in G contains $O(H)$ elements.

Partition Definition

Let S be a set, $S_1, S_2, \dots, S_n$ be the proper subsets of S .

$$\text{i) } \bigcup_{i=1}^n S_i = S \quad \text{ii) } \bigcap_{i=1}^n S_i = \emptyset$$

Normal Subgroup:

A subgroup $\{H, *\}$ of a group $\{G, *\}$ is called a normal subgroup, if for any $a \in G$, $aH = Ha$.
(i.e. the left- and right cosets of H in G generated by a are the same).

Note:

① $aH = Ha$ does not mean that $a * h = h * a$ for any $h \in H$, but it means that $a * h_1 = h_2 * a$ for some $h_1, h_2 \in H$.

② Every Subgroup of an abelian group is normal.

③ Trivial Subgroups are also normal.

Note:

A Subgroup $(H, *)$ of a group $(G, *)$ is a normal Subgroup iff $a^{-1} * h * a \in H$ for every $a \in G$ & $h \in H$.

State and prove Lagrange's Theorem:

Statement:

If G is a finite group and H be the subgroup of G then $|H| \mid |G|$.

proof:

Let G be a finite group

$$\therefore |G| = n$$

& H be a subgroup of G

$$\therefore |H| = m$$

$\therefore$ $\exists$ a bijection between two
left cosets of H in G .

i.e. for $a \in G$ & $b \in G$ also $beaH$
w.k.t. $aH = bH$.

Since H itself is a left coset
of H i.e. $eH = H$.

So every left coset of H in G
contains exactly $|H| = m$
elements.

& The distinct left cosets
of H form a partition of G .

Since G is finite so, the number of left cosets of H in G is finite.

Let $H, a_2 H, a_3 H, \dots, a_k H$ be the distinct left cosets of H in G .

$$G = H \cup a_2 H \cup a_3 H \cup \dots \cup a_k H$$

$$O(G) = O(H) + O(a_2 H) + O(a_3 H) + \dots + O(a_k H)$$

$$= O(H) + O(H) + \dots + O(H)$$

$$= k(O(H))$$

$$\therefore k = \frac{O(G)}{O(H)}$$

$$\therefore O(H) / O(G)$$

P.T. Intersection of 2 normal subgroups of a group is a normal subgroup in G but the union need not be.

proof:

Let H & K be the two normal subgroups in G .

$\Rightarrow H$ & K are subgroups of G

$\Rightarrow H \cap K$ is a subgroup of G .

$\Rightarrow x \in H \cap K$

$\Rightarrow x \in H$ & $x \in K$

Since H & K are normal in G ,

$\therefore a x a^{-1} \in H$ & $a x a^{-1} \in K$ for $a \in G$

$\therefore a x a^{-1} \in H \cap K$ for $a \in G$

$\therefore x \in H \cap K$

$\therefore H \cap K$ is a normal subgroup

but for union,

Let $Z = \{0, \pm 1, \pm 2, \dots\}$

be the group and also abelian

$\therefore H = \{0, \pm 2, \pm 4, \dots\}$ &

$K = \{0, \pm 3, \pm 6, \dots\}$ be
the two normal subgroups.

but HOK is not closed.

$\therefore HOK$ is not a subgroup
& it is not a normal subgroup.